

* Lab 4 : Nessus Vulnerability Scanner

Objectives

1. Install / configure Nessus
2. Run basic, credentialed, and policy-based scans
3. Perform vulnerability analysis, prioritization, and remediation.
4. Export and present scan report

Lab Env Setup

- Analyst VM : Linux or Windows with Nessus
- Target VMs : Windows Server, Ubuntu Server, Metasploitable 2/3, DVWA
- Network : Isolated NAT or host-only

=> Step 1 - Nessus Installation

1. Download Nessus package (Linux example):

```
sudo dpkg -i Nessus-<version>-deb1am6
```

```
sudo systemctl start nessusd
```

2. Open browser: `https://localhost:8834`

3. Create admin account and activate license Key.



=> Step 2 - Basic Network Scan

1. Go to Scans -> New Scan

2. Choose Basic Network Scan

3. Configure :

- Name : Lab Scan 1
- Targets : IP range of lab VMs

4. Save and Launch



=> Step 3 - Credentialed Scan

1. Create new scan -> Advanced Scan
2. Add credentials: SSH for Linux, SMB/WinRM for Windows.
3. Run scan

Deliverable: Additional findings such as missing patches, weak passwords.

⇒ Step 4 - Policy Creation

1. Clone Basic Network Scan policy
2. Modify plugin families : enable web app checks.
3. Save and run.

Deliverable : Policy file and scan report.

⇒ Step 5 - Report Analysis

1. Open Scan results
2. Identify top 5 critical vulnerabilities.
3. Map to CVEs
4. Export PDF & CSV reports.

Deliverable : Executive summary and technical annex.



=> Step 6 - Vulnerability Remediation

1. Choose one Windows and one Linux host.
2. Apply patches or disable weak services.
3. Rescan to verify remediation.

Deliverable: Before / after comparison.